Signing Shadow: Redshift

Cryptography



I reviewed the challenge files, which provided a Schnorr signature transcript, 24-bit timestamp offsets, an AES-GCM ciphertext, and group parameters p, q, g.

Checking the group order in data.json revealed that while q was a 160-bit prime, group modulus p was only 165 bits, rendering the discrete logarithm problem solvable via Number Field Sieve.

The challenge provided precomputed logarithmic values log(g) and log(pub) within the subgroup. I computed private scalar $x \equiv \log(pub) \times \log(g)^{-1} \bmod q$, verified $g^x \equiv pub \bmod p$, and derived the AES-GCM decryption key.

Finally, I tested the recovered flag against the executable / service:

python3 Signing_Shadow_Redshift.py

Group order verified. Subgroup discrete log inverted.

Flag Captured

SecLeaf{cl0ck_drift_c4nt_h1de_a_shad0w}

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